MIS 720 STEP 2 DISCUSSION

From Beats to Hits

Predicting Spotify Song Popularity

Team B2 | Spring 2026

Agenda

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Project & Data

Spotify track popularity, working dataset

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EDA: Continuous

Distributions and class differences

3

EDA: Categorical

Levels, frequencies, genre signal

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Methods

Logistic regression, random forest, SVM

5

Results Snapshot

Test-set comparison

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Discussion

Concerns and questions for the meeting

Research Question

Can Spotify audio features help predict whether a track is “high popularity”?

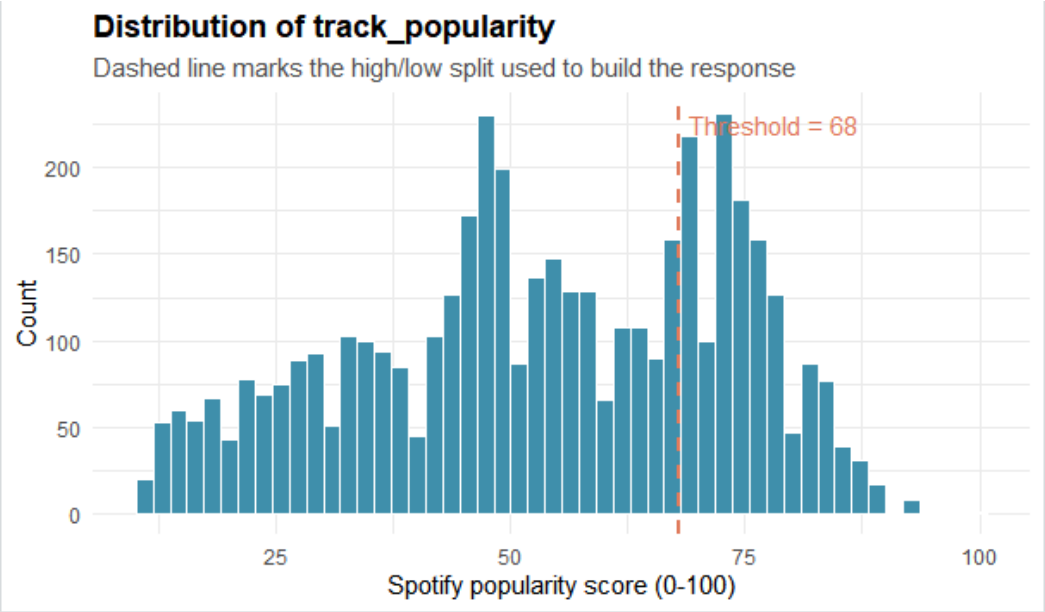
Response variable

high_popularity = 1 if track_popularity > 68, otherwise 0

Business angle: Useful for artists, labels, playlist teams, and music marketers.

Modeling angle: Binary classification with interpretable and nonlinear models.

Sub-question: How much of the predictive signal comes from intrinsic audio features versus the playlist-assigned genre?



The high/low split is based on the popularity threshold of 68.

WHAT WE ANALYZED

Data Source & Working Dataset

4,494	29.5%	70.5%	14 / 29
unique tracks after cleaning	High class	Low class	predictors used in modeling

Predictor groups

Continuous audio features	danceability, energy, loudness, valence, acousticness, instrumentalness, speechiness, liveness, tempo, duration_ms
Categorical features	key, mode, time_signature, playlist_genre
Excluded identifiers	track_id, track_name, artist, album and other text metadata retained only for traceability
Train-Test Split	5-fold cross validation with a held-out test set containing 20% of cleaned dataset

Data prep already completed

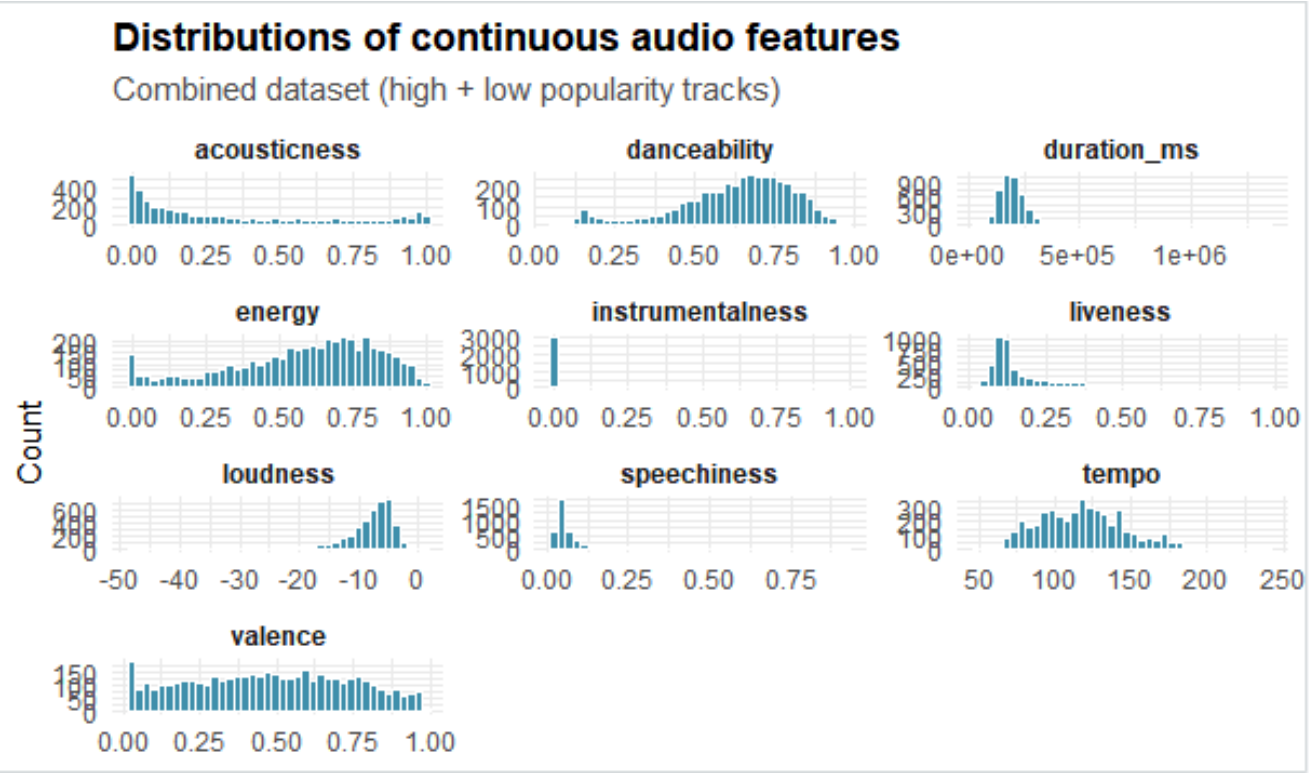
Merged high- and low-popularity CSVs.

Removed duplicates by retaining the highest-popularity row per track_id.

Dropped one row with missing audio features.

Created binary response variable from threshold 68.

EDA: Continuous Audio Features



What we learn

Danceability, energy, valence, and tempo look reasonably distributed for modeling.

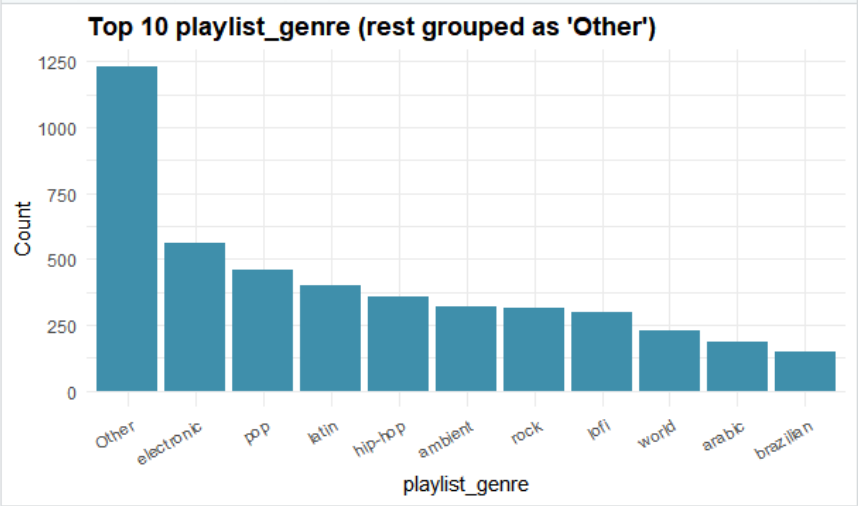
Acousticness and instrumentalness are highly right-skewed: most tracks are near zero.

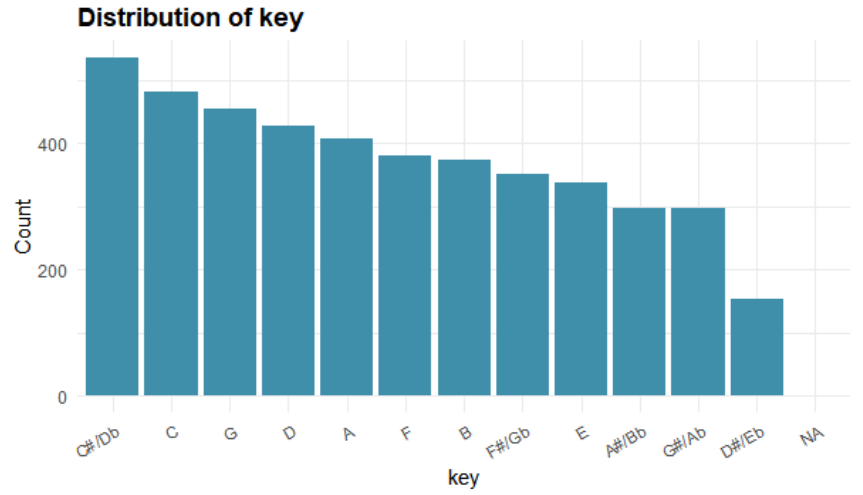
Loudness clusters near 0 dB but has a long left tail.

No distribution looks so abnormal that it blocks modeling; all numeric predictors are z-standardized for the linear and SVM models.

REQUIREMENT: LEVELS AND BAR CHARTS

EDA: Categorical Predictors





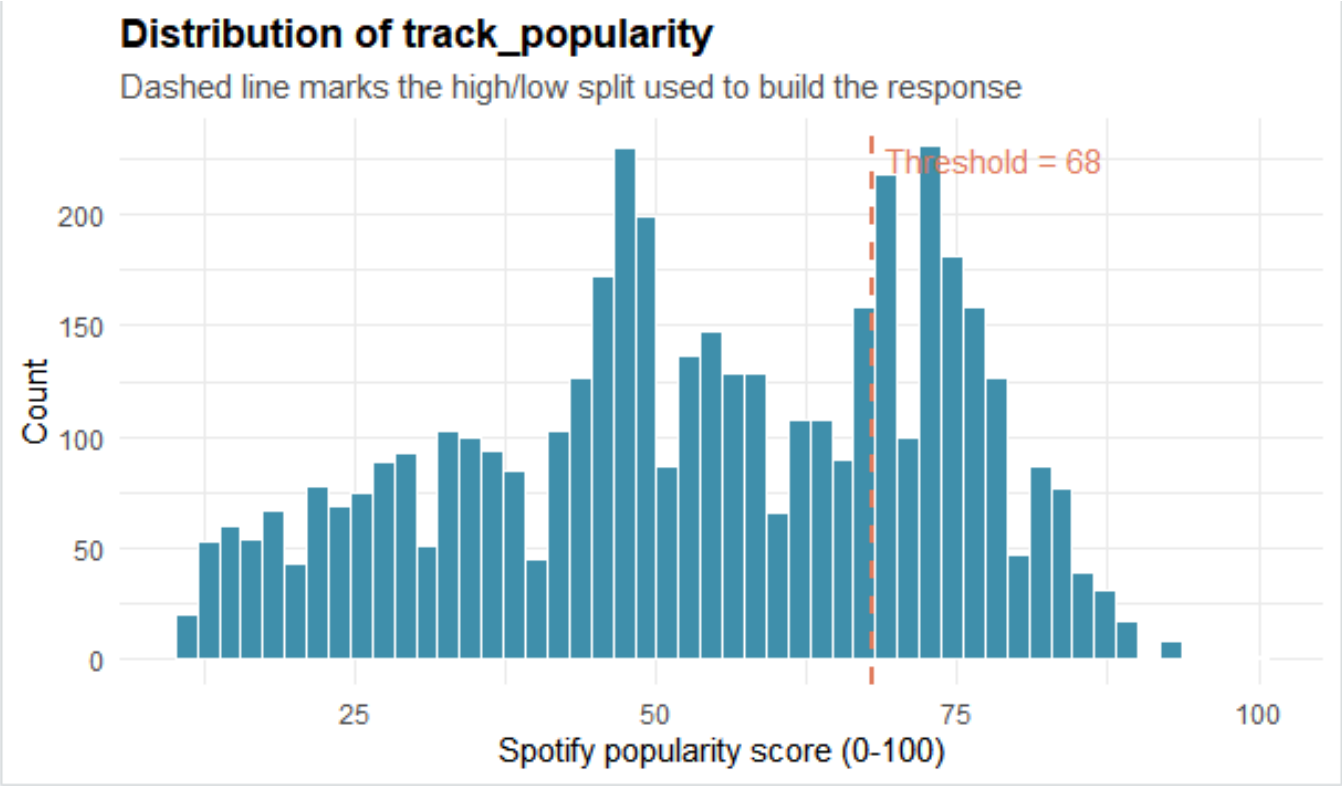
Main points

playlist_genre is high-cardinality: 35 levels. Rare levels are grouped as “Other” for modeling.

playlist_subgenre has 84 levels and overlaps with playlist_genre; we exclude it from modeling.

Musical key has 12 levels and is fairly distributed; **mode and time_signature are included as categorical predictors**.

EDA: Response Distribution & Class Balance



Response setup

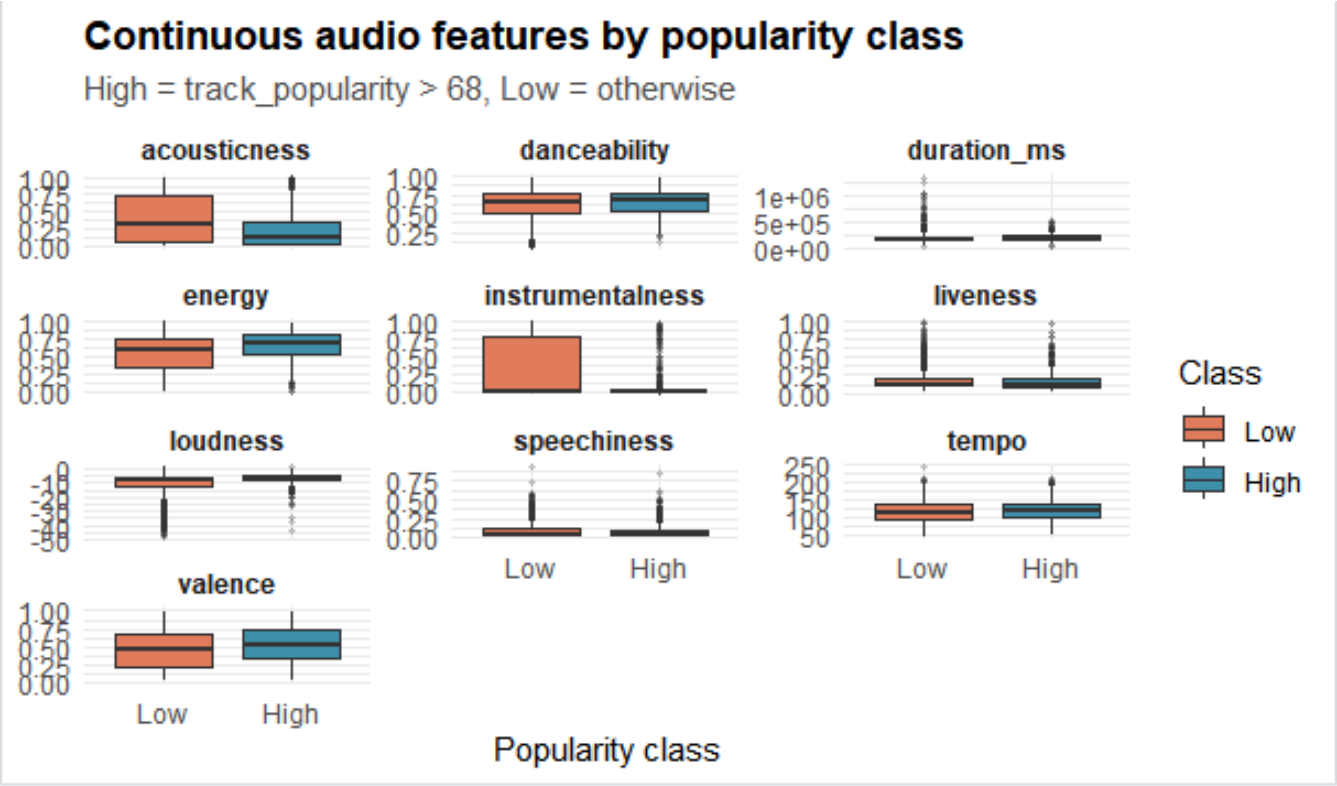
track_popularity is converted to a binary response using threshold = 68.

High = tracks above 68; Low = all others. Class balance after cleaning: 1,325 High vs. 3,169 Low.

This makes the project a classification problem, not a continuous regression problem.

REQUIREMENT: BOXPLOTS FOR CATEGORICAL RESPONSE

EDA: Continuous Predictors vs. Popularity Class



Strongest class differences

High-popularity tracks tend to be louder and more energetic.

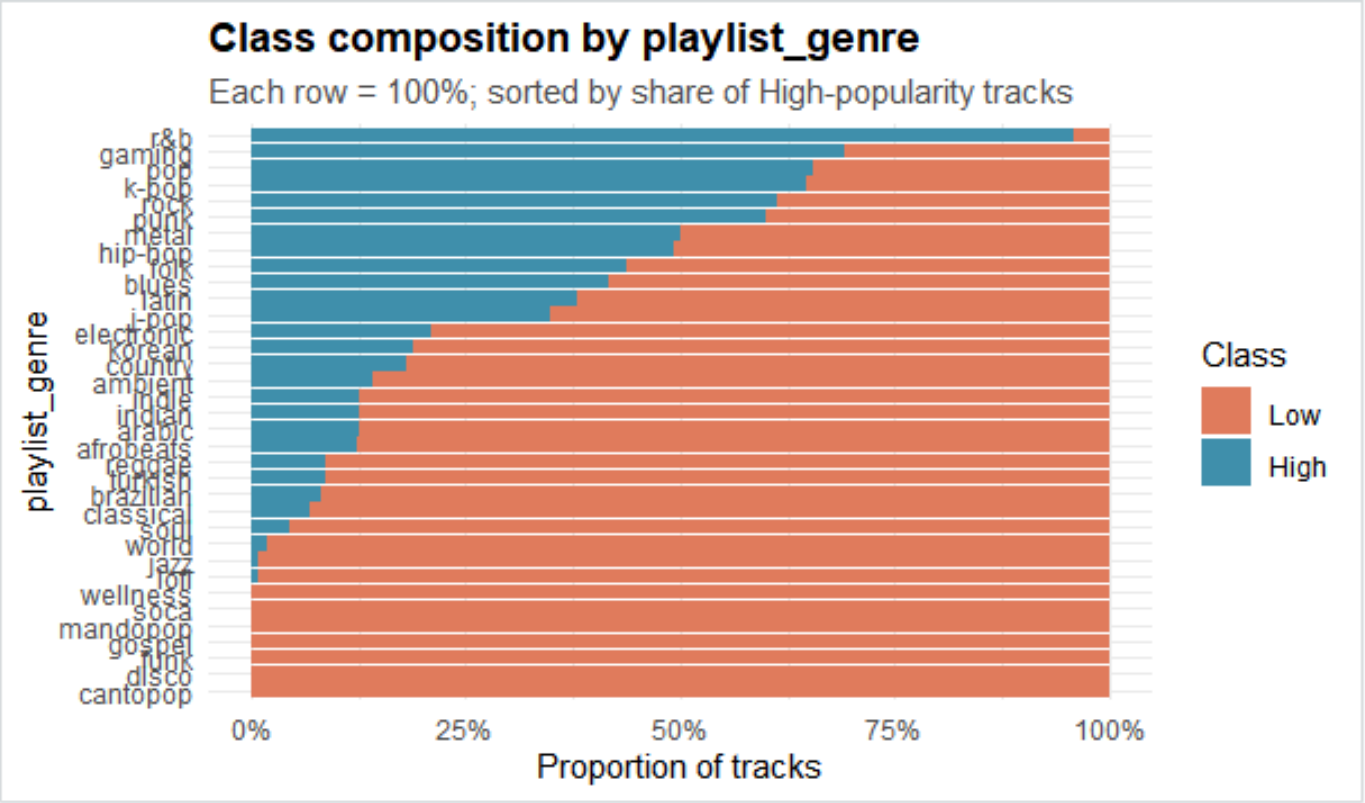
High-popularity tracks tend to have lower instrumentalness and lower acousticness.

Danceability, valence, tempo, and duration show smaller differences.

Speechiness and liveness show limited separation.

CATEGORICAL PREDICTOR RELATIONSHIP

EDA: Genre vs. Popularity Class



Why this matters

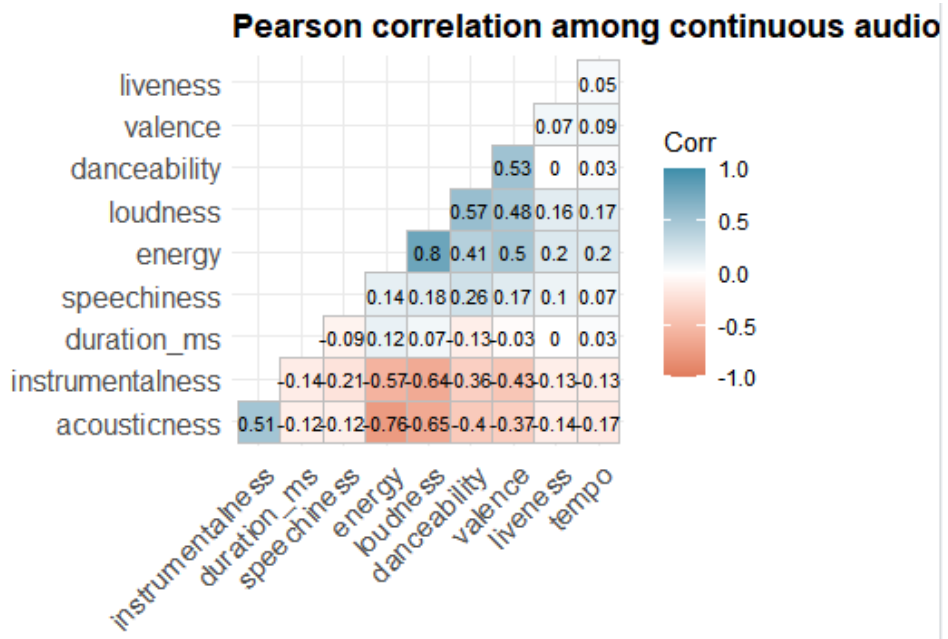
High-popularity share differs strongly across playlist_genre.

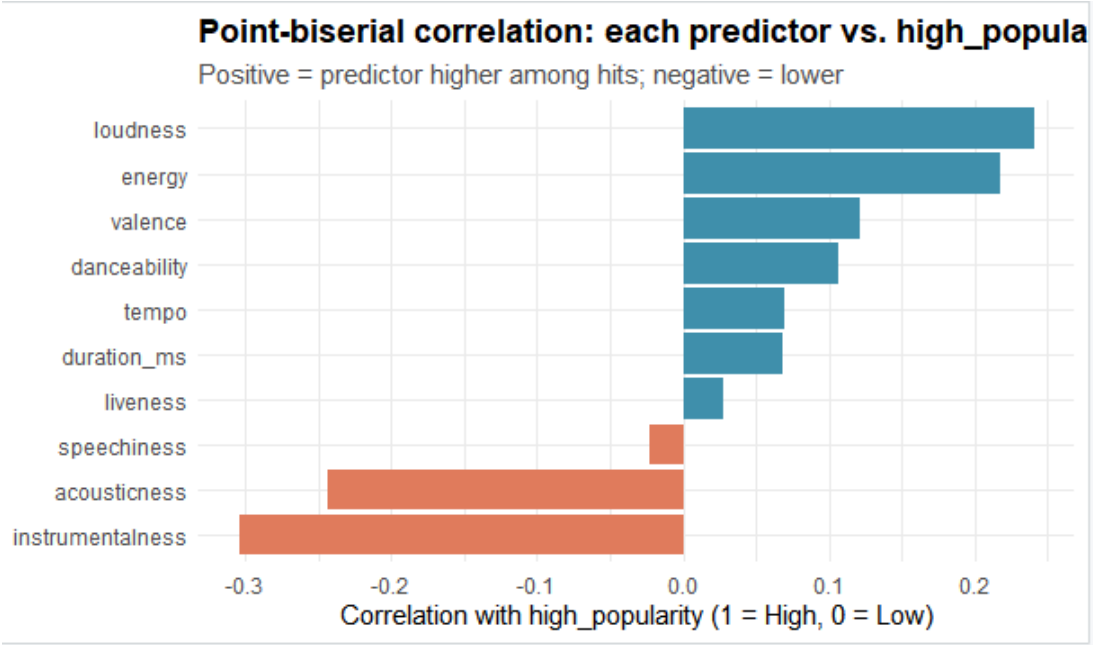
Genres such as pop, hip-hop, R&B, rock, and gaming have higher High-class shares.

Several ambient/world/jazz/lofi-type genres show very low High-class shares.

This motivates the audio-only vs. audio+genre model comparison.

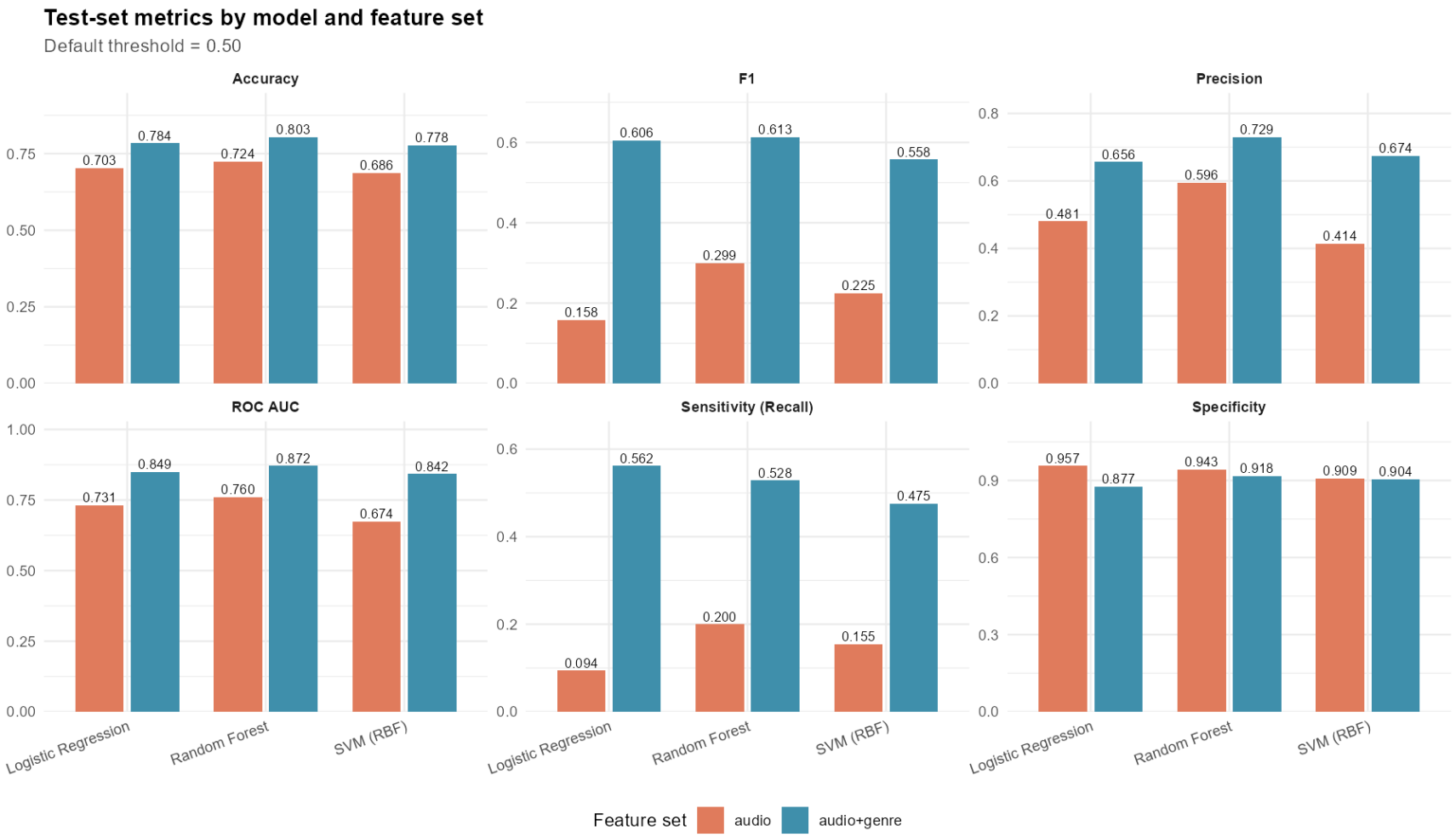
EDA: Correlation & Feature Signals





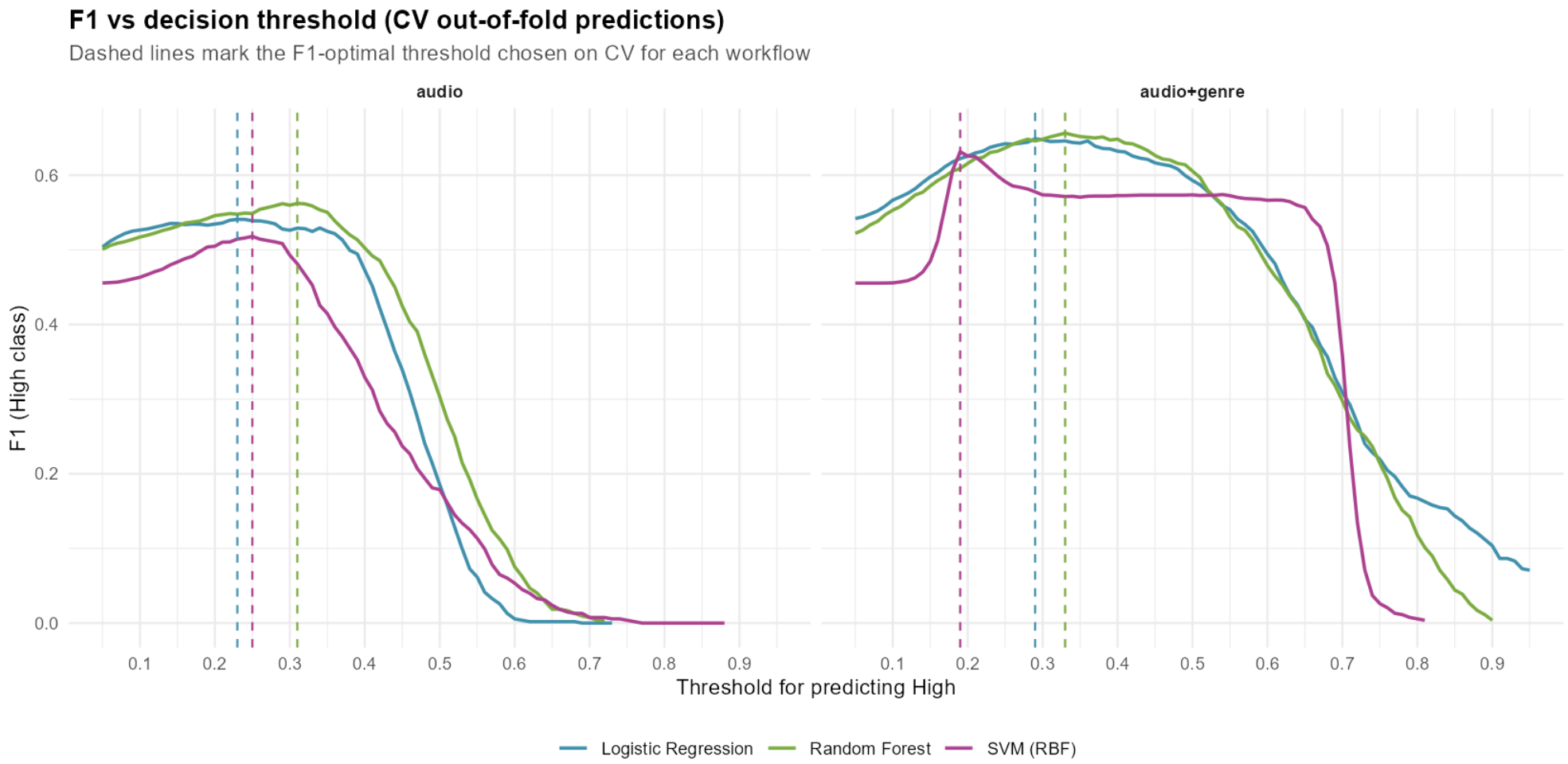
Key takeaways: energy and loudness are highly correlated (r = 0.80); instrumentalness and acousticness are negatively associated with high popularity; loudness and energy are positively associated with high popularity.

Modeling pipeline & test-set results



Three classifiers (logistic regression, random forest, SVM) trained on two feature sets (audio vs audio+genre); 5-fold CV on 80% train, evaluation on held-out 20% test, threshold tuned per workflow on CV out-of-fold predictions. Random forest on audio+genre is best (AUC 0.872, F1 0.690); adding playlist_genre lifts AUC by 0.07–0.17 across all three models.

Modeling pipeline & test-set results - Continued



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Questions for Discussion

1. **Threshold reporting** — should the final report focus on the CV-tuned thresholds (our preference) or the conventional 0.5?
2. **SVM underperformance** — keep all three models for comparison, or drop SVM and add a different method (neural network)?
3. **Fourth model** — add neural network, or stay at three?